

Azure SRE Agent

A 90-minute customer / partner session on agentic operations, governed autonomy, and the Contoso Trek demo.

Azure Container Apps

Azure Monitor

GitHub

SRE Agent

90-minute flow

Time	Segment	Outcome
0–10	Opening and on-call problem	Shared urgency
10–30	What it is + architecture	Operating model
30–50	Governance, benefits, pricing, limits	Adoption confidence
50–80	Live demos 1–4	Evidence in action
80–90	FAQ and CTA	Agree next steps

The on-call problem

Too much signal, too little context, and too many late-night decisions made under pressure.

Why incidents still take too long

Context switching

Telemetry, resource state, code, work items, runbooks, and chat history live in different tools.

Manual correlation

Engineers reconstruct topology, recent changes, and prior fixes while the incident clock runs.

Approval anxiety

Teams want automation, but not unbounded production changes.

Knowledge loss

Lessons from one incident rarely become searchable guidance for the next one.

What Azure SRE Agent is

A Microsoft-managed agentic AI service for cloud operations, incident response, and reliability workflows.

Product in one sentence

Azure SRE Agent connects to Azure resources, observability tools, incident platforms, and source-code repos so teams investigate and respond from one grounded workspace.

Automate incidents

Alert → query → correlate → root cause → mitigation proposal or action.

Automate workflows

Scheduled tasks run proactive checks and compliance sweeps.

Investigate & advise

Natural-language Q&A with connected evidence and citations.

Timeline and maturity

May 2025

Public preview announced at Microsoft Build.

Mar 2026

General availability with new capabilities.

Mid 2026

Azure Connector Namespace public preview.

Aug 2026

30-day trial GA, VNet GA, Live Reports preview.

Newest capabilities to know

Live Reports (preview)

Saved, reusable, auto-refreshing operational dashboards created from chat.

VNet integration (GA)

Route outbound traffic through customer VNet, NSG, DNS, and firewall.

30-day trial (GA)

Always-on charges waived for new customers, up to three agents; consumption still applies.

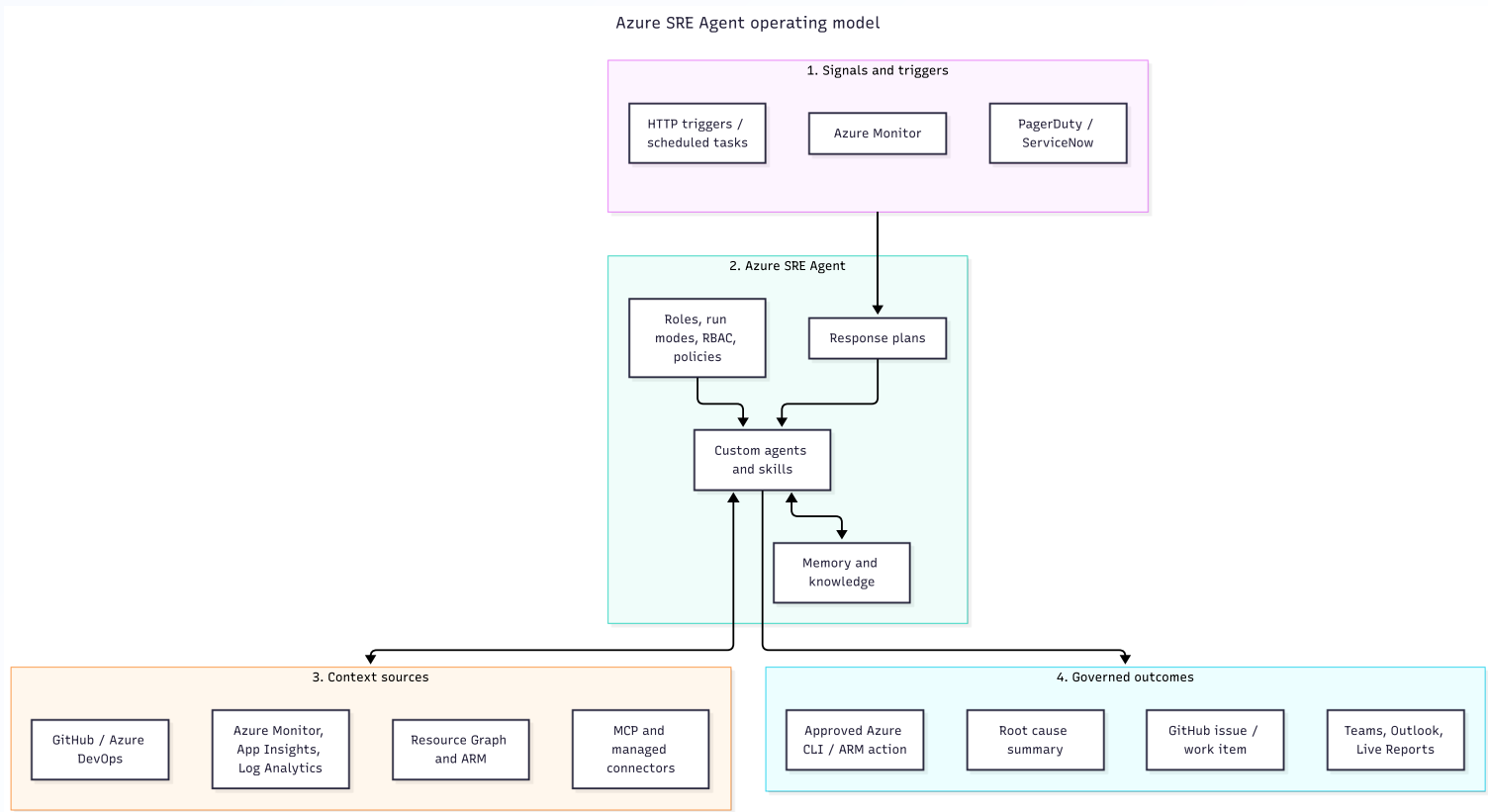
Connector Namespace (preview)

Managed MCP server hosting; early catalog includes Azure SQL, Cosmos DB, GitLab, Jira, PagerDuty.

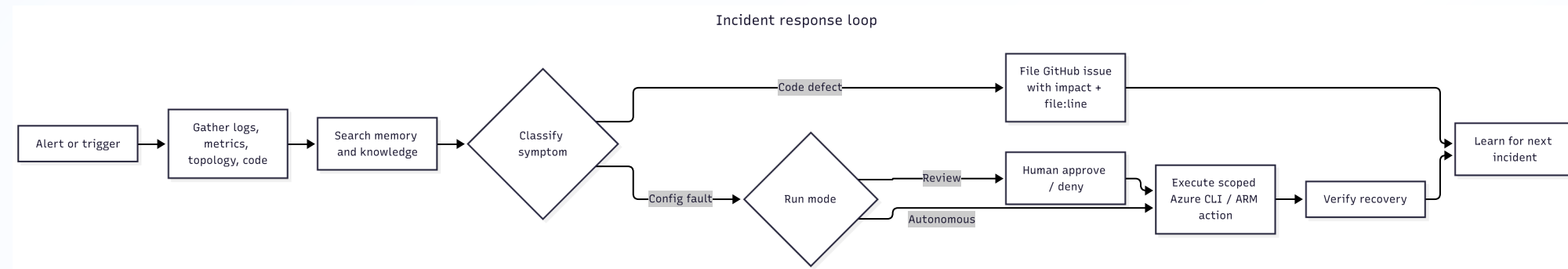
Architecture and how it works

Signals enter; context is gathered; governed agents act or advise; memory improves the next run.

Operating model



Incident response loop



Incident platforms and response plans

Platforms

Azure Monitor, PagerDuty, and ServiceNow. Only one active incident platform at a time.

Azure Monitor details

Scanner checks every 1 minute, up to 250 alerts per API call, status sync every 5 minutes, merge lookback 7 days.

Memory and knowledge

Past incidents

Automatic learning captures symptoms, working steps, root cause, and pitfalls.

User memories

Commands #remember, #retrieve, and #forget store explicit facts.

Knowledge base

Uploaded runbooks and docs become searchable, cited operational context.

Skills, custom agents, response plans

Skills

Reusable procedures with optional tools and files; auto-loaded when relevant.

Custom agents

Specialists with prompts, tools, connectors, and skills.

Response plans

Route matching incidents to the right specialist and autonomy level.

Connectors, triggers, collaboration

MCP connectors

Datadog, Splunk, New Relic, Dynatrace, Elasticsearch, GitHub; 80-tool budget; 60-second heartbeat.

HTTP triggers

Webhook endpoints for CI/CD and external monitoring systems.

Scheduled tasks

Recurring natural-language investigations that create full threads.

Teams / Outlook

Post updates to Teams or send summaries by email; managed connectors add preview governance.

Code Access and developer workflow

Code Access

Configured by PAT with `PUT /api/v2/repos/{name}` so the agent can search code and read files by path/branch.

GitHub MCP connector

Agent connector type `Mcp` at `https://api.githubcopilot.com/mcp/`; selected tools are exposed as `github_*`.

Selected tools

`issue_write`, `issue_read`, `search_issues`, `add_issue_comment`, `search_code`, `commits`, `file reads`, and `assign_copilot_to_issue`.

Developer hand-off

The code investigator de-duplicates, files the issue with file:line, then assigns GitHub Copilot coding agent for a draft PR.

From incident to pull request

1. SRE Agent

Correlates alert, telemetry, stack trace, source, and prior knowledge.

2. GitHub issue

Searches existing issues, avoids duplicates, and writes impact + stack + file:line.

3. Copilot coding agent

`github_assign_copilot_to_issue` opens a draft PR for human review.

Human control point: the PR is reviewed and merged by the team; SRE Agent observes post-deploy signals to verify recovery.

Governance and security

Autonomy is useful only when the blast radius is explicit and auditable.

Governance stack

Four-layer governance stack



Roles and separation of duties

Role	Best-fit responsibility
SRE Agent Reader	View threads, logs, and incidents.
SRE Agent Standard User	Chat, run diagnostics, request actions, scheduled tasks, knowledge uploads.
SRE Agent Author	Create custom agents and response plans; configure incident management.
SRE Agent Administrator	Approve actions, manage connectors, delete resources, full control.

Run modes and RBAC are different

Run mode

Review is default; Administrator approves/denies Azure infrastructure writes. Autonomous executes immediately within permissions.

Permission level

Reader grants read diagnostics and OBO fallback; Privileged grants resource-type contributor roles on selected resource groups.

Data, network, audit posture

Data use

Microsoft does not use customer data to train AI models; data is tenant/subscription isolated.

Residency

Conversation/content data is stored and processed in the selected Azure region.

Network

VNet integration is GA and optional: Unrestricted, Limited, or Azure VNet.

Benefits

Faster investigation, lower toil, safer repeatability — with cited Microsoft and customer proof points.

Quantified proof points

3,000+

Internal Microsoft services scaled on Azure SRE Agent.

1.5M+

Incidents processed internally at Microsoft.

50%

Up to half resolved autonomously in Microsoft internal usage.

Customer signal: InEight reported 80% reduction in incident investigation time and 80% reduction in build-failure triage time.

Pricing and cost controls

AAUs make agent cost visible: fixed always-on plus variable active-flow consumption.

Pricing model

Always-on flow

4 AAUs per agent-hour from creation until deletion. Stopping pauses active flow, not always-on billing.

Verified AAU unit price

Retail Prices API, queried 2026-10-03: East US / East US 2 **\$0.10**; France Central / Sweden Central **\$0.11**; West Europe **\$0.14** effective 2026-10-01; France Central **EUR 0.10**.

Quote discipline: prices vary by region and currency; confirm current values in the Azure pricing calculator for customer quotes.

Active-flow AAU rates per 1M tokens

Model	Input	Output	Cache read	Cache write
Claude Opus 4.6	100	500	10	125
GPT 5.3 Codex	35	280	3.5	0
GPT 5.2	35	280	3.5	0

Official examples: quick question $\approx$ 3.8 AAUs (Claude) / 1.3 (GPT 5.3 Codex); incident investigation $\approx$ 35.3 / 11.7; full remediation $\approx$ 86.5 / 30.1.

Worked monthly estimate

Always-on floor

4 AAU $\times$ 730 h = **2,920 AAU/month** $\approx$ \$292 in East US or \$321 in France Central.

20 investigations/month

GPT-5.3 Codex example: $11.7 \times 20 \approx$ **234 AAU** $\approx$ \$23 East US / \$26 France Central.

Claude investigation mix

Claude Opus example: $35.3 \times 20 \approx$ **706 AAU** $\approx$ \$71 East US / \$78 France Central.

Small-team monthly shape

1 agent + 20 GPT investigations + 10 GPT remediations + 100 quick questions $\approx$ **3,585 AAU** $\approx$ \$359 East US / \$394 France Central.

Hours comparison: break-even hours = monthly agent cost $\div$ your loaded on-call engineering cost/hour. At an illustrative \$100/hour, the small-team East US shape breaks even at about 3.6 saved hours/month.

Budget cap and trial

Monthly active-flow limit

Configurable 500 to 1,000,000 AAUs. Hitting it pauses chat/actions; always-on billing continues.

30-day trial

Always-on charges waived for up to 3 new agents. Active-flow charges still apply; deleted agents still count.

Limits and quotas

Adoption planning needs the sharp edges on the table early.

Selected published limits

Area	Limit / fact
Regions	22 confirmed supported Azure regions; one deployment region; cannot change after creation.
MCP tools	80 tools per agent; 60-second health heartbeat; new tools detected within 5 minutes.
Knowledge	.md/.txt only; max 50 MB per file; up to 1,000 files per custom agent instance.
GitHub OAuth	10 OAuth tokens per agent.
Incidents	One active incident platform per agent.

Preview / verify items

- **Preview:** Managed connectors, Connector Namespace, Live Reports, and control/data-plane REST APIs have preview caveats in current docs.
- **Verify before automation:** exact API version, service-principal auto-admin behavior, and concurrent-thread quotas.
- **Quote discipline:** AAU prices are verified for the listed regions, but prices vary by region/currency; confirm current values in the calculator for customer quotes.

Live demo runway

Contoso Trek: an outdoor-gear storefront on Azure Container Apps with Azure SRE Agent as operator.

Demo 1 — deploy from GitHub Actions

Show the deployment story: source, workflow, Bicep, images, agent resource, and smoke test.

Demo 1 talk-track

Show

- `.github/workflows/deploy.yml`
- `scripts/deploy.sh`
- Bicep modules under `infra/`
- GitHub Actions run summary

Say

The agent is deployed and configured like any Azure resource: versioned, reviewed, and repeatable.

```
bash scripts/deploy.sh
bash scripts/configure-agent.sh
bash scripts/smoke-test.sh
```

Agent config as code

Response plans

app-exception → code-investigator
availability → platform-operator

Least privilege by tool grant

Code investigator: no Azure write tools. Platform operator: Azure CLI write but no terminal/GitHub. Read-back verification reports
phantom=none .

```
sre-config/agent-config.json  
sre-config/skills/*.md  
sre-config/instructions/*.md  
sre-config/knowledge-base/*.md
```

Demo 2 — code defect to GitHub issue

A Climbing product detail throws HTTP 500; the agent identifies the source defect and files an issue.

Demo 2 steps and expected outcome

Run

```
REQUESTS=20 bash scripts/trigger-errors.sh
```

Wait 5–10 minutes for the alert, or use `scripts/ask-agent.sh` to skip alert latency.

Expected

- Agent follows AppExceptions and stack trace.
- Searches existing issues to avoid duplicates.
- Files GitHub issue with impact, stack excerpt, file:line, suggested fix.
- Hands off to Copilot coding agent with `github_assign_copilot_to_issue` for a draft PR.

Demo 3 — config fault to autonomous fix

A bad Container App environment variable causes 503s; the agent restores known-good configuration and verifies recovery.

Demo 3 steps and expected outcome

Run

```
BAD_INVENTORY_BACKEND=cosmosdb-prod bash scripts/break-config.sh
```

The availability alert fires in about 5–10 minutes.

Expected

- Finds `CONFIG_ERROR` traces.
- Correlates Activity Log / revision change.
- Runs `az containerapp update ... INVENTORY_BACKEND=builtin`.
- Verifies `/health/ready` and status return green.

Demo 4 — chat, scheduled task, Q&A with the agent

Use the agent as a reliability analyst before and after alerts.

Demo 4 prompts

```
bash scripts/ask-agent.sh "Summarize Contoso Trek health and recent errors. Cite evidence."  
bash scripts/ask-agent.sh "Create a concise readiness report for ca-trek-api and ca-trek-web."  
bash scripts/ask-agent.sh "What recurring scheduled task would you recommend for this workload?"
```

Show at sre.azure.com: chat thread, evidence links, tool calls, and scheduled task management.

Getting started

Pilot safely: choose scope, seed knowledge, measure outcomes, then expand autonomy.

Prerequisites and deployment choices

Azure prerequisites

Microsoft.App provider registered; Owner or Contributor + User Access Administrator; supported region; monitoring available.

Automation paths

Portal for evaluation; Bicep / ARM / Terraform AzAPI; Azure Developer CLI recipes; GitHub Actions for Contoso Trek.

Contoso Trek repo quick start

```
# validate/deploy/configure/smoke-test
bash scripts/deploy.sh
bash scripts/configure-agent.sh
bash scripts/smoke-test.sh

# break/fix scenarios
REQUESTS=20 bash scripts/trigger-errors.sh
BAD_INVENTORY_BACKEND=cosmosdb-prod bash scripts/break-config.sh
bash scripts/fix-config.sh
```

Default environment: `demo`; default location: `francecentral1`; resource group: `rg-sre-agent-demo-demo`.

Adoption roadmap

Crawl

Reader + Review. Connect telemetry, import runbooks, ask questions, measure triage time.

Walk

Review on scoped resource groups. Add response plans, scheduled tasks, GitHub/ADO issue flow.

Run

Autonomous for narrow tasks. Allow safe, reversible repairs on non-prod or tightly scoped production resources.

Competitive landscape: fair positioning

Strong Azure-native fit

Deep native access to ARM, Resource Graph, Azure Monitor, App Insights, Log Analytics, AKS, Container Apps, App Service, and Functions.

Connector-mediated multi-tool fit

MCP, HTTP triggers, and managed connectors reach non-Azure telemetry and SaaS workflows.

Competitor strengths

PagerDuty: incident orchestration; Datadog: telemetry-native AI; cloud-neutral agents: multi-cloud-first posture.

Azure differentiator

Governed autonomy + Azure control plane + IaC-managed config + persistent memory.

Call to action

Pick one high-signal service, two recurring incident types, and a Reader + Review pilot.

Week 1

Deploy agent, connect Azure Monitor, upload runbooks.

Week 2

Configure response plans and GitHub/ADO workflow.

Week 3–4

Measure MTTR, toil, false positives, and candidate autonomous repairs.

Resources

- Docs: <https://learn.microsoft.com/en-us/azure/sre-agent/overview>
- Product page: <https://azure.microsoft.com/en-us/products/sre-agent/>
- Pricing model: [pricing-billing doc](#)
- Retail Prices API: <https://prices.azure.com/api/retail/prices>
- Portal: <https://sre.azure.com>
- Contoso Trek assets: `infra/`, `scripts/`, `sre-config/`, `src/`
- Session materials: `docs/session/`

Sources used in this deck

- Microsoft Learn Azure SRE Agent docs: overview, pricing, evaluation, roles, run modes, permissions, memory, connectors, incidents, response plans, scheduled tasks, triggers, hooks, policies, network, privacy, regions, API, IaC, Live Reports.
- Azure Updates RSS: May 2025 preview, Mar 2026 GA, Aug 2026 trial / VNet / Live Reports announcement.
- Microsoft Developer blog: internal scale and InEight proof points.
- Microsoft Azure SDK Dev Blog: Azure Connector Namespace.
- Local demo contract and repository scripts/configuration.